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...Go Beyond Realms

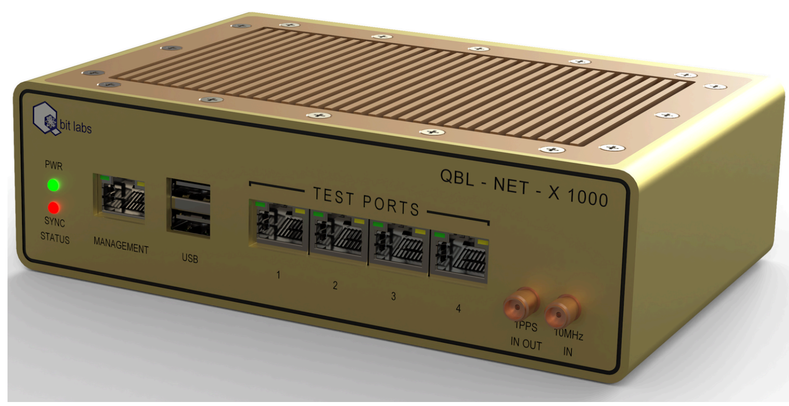
QBL-NET-X1000 COMPACT TSN NETWORK TESTER

The QBL-NET-X1000 Compact Network Tester is a portable, lab-grade test instrument designed for Layer 2–3 performance validation, deterministic networking (TSN/DetNet) compliance testing, and advanced traffic generation. The platform provides four simultaneous copper Ethernet test ports.

Portable 4-Port Layer 2 / Layer 3 / TSN Performance & Traffic Generation Solution

Features & Benefits

- Latency Measurement Accuracy: Up to ±3.2 ns (hardware timestamping).
- Synchronization Precision: Sub-microsecond, multi-domain support.
- Traffic Rates: Line-rate up to 1 Gbps per port; support for 100/1000BASE-T.
- Layer 2–3 performance and conformance testing
- Integrated L2/L3 traffic generation
- 4-port simultaneous traffic generation and analysis
- TSN compliance testing (network & component level)
- Deterministic latency, jitter, and packet-loss validation
- Hardware Based Checksum calculation
- Timestamping: Hardware-level (ns resolution).
- External Sync: GPS/1PPS input for reference clock
- RFC-based benchmarking and service activation



Target Applications

🔀 TSN Switch & End-Node Validation
🏭🚗 Industrial & Automotive Ethernet Testing
⏱️ Deterministic Networking Verification
📊 Enterprise Network Benchmarking
🧪📡 Field Acceptance Testing

Why Choose Us?

🚀 Rapid integration, faster time-to-market 📘 Clean, verified, documented RTL 🤝 Direct access to experts

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Technical Specifications

1. Protocol Support

Specifications	Details
Time Synchronization	• gPTP master/slave/monitor (IEEE 802.1AS-2011/2020/Rev) • IEEE 1588v2 (PTP) • Multi-domain emulation and redundancy • External clock reference sync
Traffic Scheduling	• Time-Aware Shaper (IEEE 802.1Qbv) • Credit-Based Shaper (Qav) • Asynchronous Traffic Shaping (IEEE 802.1Qcr) • Stream Reservation Protocol (IEEE 802.1Qat) • Forwarding & Queuing (IEEE 802.1Qav)
Redundancy & Reliability	• FRER (IEEE 802.1CB) - Seamless redundancy
Ingress Policing	• Per-Stream Filtering & Policing (IEEE 802.1Qci)
Frame Preemption	• Express/preemptible traffic (IEEE 802.1Qbu)
Routing & Switching Protocols	• BGPv4/BGPv6 • IS-ISv4/IS-ISv6 • OSPFv2/OSPFv3 • PIM-SM/PIM-SSM • BFD - STP/RSTP/MSTP • LACP/Static LAG/Protocols over LACP (PoLACP) • IGMP/MLD - DCBX/LLDP - FabricPath/TRILL
MPLS & VPN	• RSVP-TE P2P/P2MP • LDP/LDPv6/mLDP • L3 MPLS VPN/6VPE/6PE • L2 LDP VPN/PWE/VPLS • BGP VPLS/VPWS (RFC3107) • EVPN/PBB-EVPN • Multicast VPN • NG MPLS OAM • MPLS over GRE/UDP

Broadband & Authentication	• PPPoE/L2TPv2 • DHCPv4/DHCPv6 (incl. over EoGRE) • IPv6 Autoconfiguration (SLAAC) • Bonded GRE - 802.1x - ANCP
Industrial & OAM	• CFM (IEEE 802.1ag) • ITU-T Y.1731 • FCoE Forwarder

2. Traffic Generation Capabilities

Specifications	Details
Frame Generation	• Minimum Frame Size: 64 Byte • Maximum Frame Size:1500 Byte • Maximum Jumbo Frame Size:9500 Byte • Frame Length Control: Fixed, increment by user-defined step, weighted pairs, uniform, repeatable random,
TSN Traffic Generation	• Per-port and per-stream configuration: Rate, frame size, payload, burst, VLAN tagging, priority (PCP), stream ID. • Customizable traffic mixes: Scheduled, credit-based, asynchronous, preemptible/express, Best-effort. • Realistic mixed environments: TSN + non-TSN endpoints, multiple domains/connections. • Trackable streams with real-time statistics: Per-stream Tx/Rx counters, latency (min/avg/max), jitter, loss.
Layer 2 Traffic Generation	• Unicast, multicast, broadcast • VLAN, QinQ, PCP - CBR, bursty, back-to-back traffic
Layer 3 Traffic Generation	• IPv4 and IPv6 • UDP, TCP, ICMP • DSCP / ToS marking • Multi-flow background traffic
RFC & Service-Based Profiles	• RFC 2544: Throughput, Latency, Frame loss and Back-to-back frames • ITU-T Y.1564: Service configuration test, Service performance test and Multiple service instances • RFC 6349: TCP throughput and efficiency testing
Error Generation	• Checksum Error • Impairment insertion: Delay, jitter, loss, duplication, mis-ordering.

3. Hardware Interfaces

- **Form Factor:** Compact rugged enclosure (approx. 20x12x6 cm, <1.2 kg).
- **Ports:** 4x configurable test ports RJ45 for 10/100/1000BASE-T.
- **Power:** AC-powered via universal adapter (100–240 V, 50/60 Hz); low-power fanless design; optional DC input for bench flexibility.
- **User Indicators:** Status LEDs (power, link/activity, test status); no display.
- **Environmental:** IP65/67 dust/water resistance, MIL-STD-810H drop/shock/vibration, operating temp -10°C to +55°C.
- **Dedicated management** Ethernet port and USB

4. Conformance Testing

QBIT TSN Conformance	Coverage Area / Scenarios	Number of Scenarios
IEEE 802.1AS-2020	**Timing & Synchronization** • Sequence Id validation • Pdelay request interval validation • Announce message validation • Message packet format validation • Timing & Synchronization for TSN • Domain number validation • PTP Message Header Validation	22
IEEE 802.1Qbv	**Time Aware Shaper** • Verification of Gate Open & Gate Close States • Verification of queueMaxSDU • Verification of minimum & maximum cycle time • Verification of transmission window • Verification for Window Violation • Verification for Configuration Update over Netconf	23
IEEE 802.1Qci	**Per Stream Filtering & Policing** • Verification of "GateClosedDueToInvalidRx" scenario • Verification of "GateClosedDueToOctetsExceeded" scenario • Verification of "StreamBlockedDueToOversizeFrame" scenario	15
IEEE 802.1CB	**Frame Replication & Elimination for Reliability** • Verification of Stream Identification Function • Verification of Stream Replication • Verification of Stream Recovery Function	18

Avnu TSN conformance	Coverage Area	Specific Scenarios / Test Cases	Total Scenarios
IEEE 802.1AS-2020	Timing & Synchronization (Bridges and Endpoints)	• Common Minimal Time Aware System,Site Sync State Machine Validation, Port Sync Receive State Machine Validation, Clock Slave Sync State Machine Validation, Best Leader Clock Algorithm Validation, Port Announce Receive State Machine Validation, Port Announce Information State Machine Validation, Port Role Selection State Machine Validation, Clock Master State Machine Validation, Media-independent Leader Validation, Port Announce Transmit State Machine Validation • Media Dependent • Full Duplex • Point to Point Link Validation - Sync Receive State Machine Validation, Sync Send State Machine Validation, Pdelay Req State Machine Validation, Pdelay Resp State Machine Validation • gPTP Management • Avnu Specific Test Cases for gPTP Signaling Message Conformance • Bridge Specific Test Cases	153 + 11 (Avnu/Bridge specific)
IEEE 802.1Qbv	Time Aware Shaper (Both for Bridges and End Points)	· Verification of transmission & rejection behavior for bridges and end points · Verification of Cycle Time · Verification of Cycle time duration not equal to control list duration · Dynamic Schedule Changes · Validate maximum dynamic schedule change request to schedule change interval	26

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IEEE 802.1Q	Forwarding & Queuing of Time Sensitive Streams (Only Bridges)	· Non-SR Traffic Bandwidth is Work Conserving · One SR class, One Ingress, · One Egress Algorithm Validation · One SR class and Line · Rate interfering traffic · Maximum Reservation check · Minimum Reservation and FDB check · AVB Boundary ports properly regenerate per SR class	9
IEEE 802.1Qbv	Time Aware Shaper (For Bridges) – Industrial Coverage	· Verification of transmission & rejection behavior for bridges · Verification of Cycle Time	17

Get in touch with us

Delivering Advanced IP Solutions for Next-Gen Networking

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